

CLAIM - Alpha as a 21|30 Closure Gate v38.4

Primary box formula first; ledger links explain where each factor enters.

The formula below is the primary claim, not a later numerical add-on. The rest of the note is the ledger explaining where each factor enters.

Primary claim: α_{RT} is the closure margin

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260} (3 \cdot 30 - \Delta)$$

Expanded correction

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$

Factor map

Factor	Meaning
pi	circular / half-turn geometry scale
30	C30 carrier/readout grid
1260	first full active closure
3*30	ideal three-chain C30 opening
Delta	typed closure loss/correction
3*30 - Delta	effective remaining opening
pi/(30*1260)	carrier x full-closure normalization

Numerical comparison

external check only - not input

alpha_RT	0.0072973525630485
alpha_inv	137.035999201092
CODATA inv	137.035999177(21)
diff inv	2.409e-08

The gate value is computed from the 21|30 ledger. CODATA is displayed only after the computation.

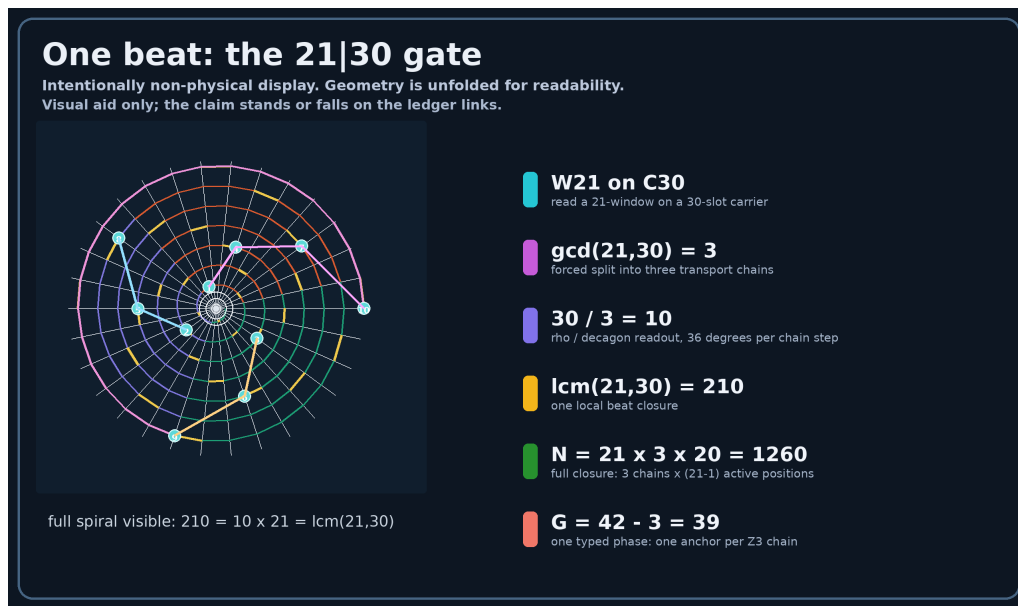
Gate ledger leading to the factors

Link	Consequence
21 30	primitive readout on C30 carrier
3 = gcd(21,30)	three-chain split; source of the 3 in 3*30
10 = 30/3	rho=10 readout scale used in Delta
210 = lcm(21,30)	one beat cell: 7*C30 = 10*21
1260 = 6*210 = 30*42	first full active closure and normalizing scale
G = 42-3 = 3*(14-1) = 39	one A/B-resolved phase reserves one anchor per Z3 chain
D = (21-1)/21 = 20/21	one boundary slot reserved inside carrier-derived 21-domain

Break condition: invalid link, non-unique gate, broken type rule, or equal-or-stricter competing grammar.

Gate detail - carrier-beat and Delta grammar

Six 210-cells split as A/B x Z3; this supports G=39 and the 21-domain for D.



Companion visualizer (not part of the claim): [RT_ALPHA_READOUT_GEOMETRY.html](https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_ALPHA_READOUT_GEOMETRY.html). The claim stands or falls on ledger links.

GitHub repo: <https://github.com/Bo-Fremling/rhythm-theory-release>

Visualizer path: https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_ALPHA_READOUT_GEOMETRY.html

Carrier-beat support lemma

Full closure contains six 210-beat cells. With $\theta(n) = 360 \times n / 1260$, each 210-boundary advances 60 degrees. A three-arm carrier therefore splits the six cells into two complementary triads.

j	$n=j \times 210$	$q=n/30$	phase	class
0	0	0	0 deg	A
1	210	7	60 deg	B
2	420	14	120 deg	A
3	630	21	180 deg	B
4	840	28	240 deg	A
5	1050	35	300 deg	B

A = {0,14,28}, B = {7,21,35} in C30 index

$1260 = 6 \times 210 = (A/B) \times Z3 \times 210$

$G = 42 - 3 = 3 \times (14 - 1) = 39$

AB=2 already counts the complementary phases; G counts one typed phase, so 42-3 not 42-6.

Each phase spans 3 beat cells; each cell has $210/30=7$ C30 slots $\rightarrow 3 \times 7=21$.

Correction grammar and type rule

Delta = AB + AB/rho - 1/(rho G) - D/(rho G C S)

Expansion: $\text{Delta} = 2 + 2/10 - 1/(10 \times 39) - (20/21)/(10 \times 39 \times 30 \times 7)$.

Type rule: each term is measured in its own domain; numerator schema is AB, AB, 1, D.

Counting measure gives numerator 1; readout-conditioned measure gives numerator D.

Active duty: one typed phase exposes $3 \times 7=21$ slots; reserve one $\rightarrow D=20/21$.

$\alpha_{\text{RT}}^{-1} = 137.035999201092$; CODATA is external comparison, not input.

Scope and break protocol

Alpha claim only. To refute: break a named gate link, the active-duty derivation, the type rule, or provide an equal-or-stricter competing primitive gate. Free recombinations do not count.

Delta as typed bookkeeping

Two port costs and two returns; same gate read on two scales.

Delta is typed bookkeeping, not four free terms

The correction is read as cost + cost - return - return. The same A/B gate is counted on two scales, and the over-count is given back on two resolutions.

$$\Delta = AB + \frac{AB}{\rho} - \frac{1}{\rho G} - \frac{D}{\rho G C S}$$

The four bookkeeping terms

Term	Value	Role	Simple image
AB	2	port cost	opening + closing of the same gate
AB/rho	2/10	inner-scale port cost	same gate measured on the rho=10 readout scale
-1/(rho G)	-1/(10*39)	coarse return	one over-counted corner is given back
-D/(rho G C S)	-(20/21)/(10*39*30*7)	fine return	return through the full 21 30 readout mask

Symbol	Value	Role	Meaning
AB	2	polarity / gate flank count	opening + closing
rho	10	internal scale	rho=30/3, internal 10-step readout scale
G	39	gap domain	G=42-3=3(14-1), one anchor reserved per Z3 chain
D	20/21	active duty	one P-ARM / boundary slot reserved inside the 21-domain
C	30	carrier base	C30 grid
S	7	beat-cell length	S=210/30, seven C30 slots per 210-cell

Interpretation: the positive terms pay for the two A/B flanks at two resolutions. The negative terms return what the gate/readout structure shows was over-counted. This does not add new numerical inputs; it names the roles already present in the typed Delta grammar.